

EE1L1 IP-1 B2-1 Project Plan

Defining Goals, Timelines, and Deliverability

Team Members: T.J. Bello, I. Givony, A. Guo,
N.T. Herijgers, S.J.H. ter Huurne, L.D. Melissant,
K.W. Smid, N.F. van Vroonhoven,
G.D. van Weelden, T.V. Wever.

Supervisor: H.E.J. Bosma

Teaching Assistant: M. Mihankhah

BSc Electrical Engineering

Faculty of Electrical Engineering, Mathematics and
Computer Science

02-12-2024



Contents

1	Introduction & Background	1
2	Project Results	2
3	Project Activities	3
3.1	Tasks Required	3
3.2	Meeting Functions	3
3.3	Work Breakdown	4
3.4	Project limits	6
4	Organization & Schedule	7
4.1	Organization & Schedule.	7
	Subgroup Schedule	9
5	Risks Analysis	12
5.1	Project Risks and Precautions	12
5.1.1	Physical Risks	12
6	Conclusion	14
	Power Supply	14
	Power Amplifier.	14
	Filters	14
7	Team Commitment Contract	16

Introduction & Background

Integrated Project 1 is a vital aspect of the Bachelor of Electrical Engineering. In the first quarter, students collaborate in teams to design and build an amplifier system for a speaker leveraging knowledge acquired in the Linear Circuits A and B course. The amplifier system will be divided into several components: three separate filters tailored to each of the distinct speaker cones, a power supply, and a power amplifier.

Group B2-1 has two lab sessions per week, during which there will be a focus on planning, building, and evaluating the amplifier system we designed. At the end of the project, our extension will be attached to a speaker and evaluated using TU Delft's speaker measurement equipment. Assessments will be based on the sound quality of the speaker, with performance comparisons made between all groups to determine which group has the most successful design.

2

Project Results

The primary objective of this Integrated Project is to design and synthesize an amplifier for a loudspeaker system consisting of three distinct speaker cones: a woofer, a mid-toner, and a tweeter. If the primary goals are achieved early, there is the potential to extend the Integrated Project by incorporating the Booming Bass Extension alongside the main amplifier. The amplifier system will be divided into several components: three separate filters tailored to each of the distinct speaker cones, a power supply, and a power amplifier.

There are a few specifications and requirements for the loudspeaker; these specifications should be matched to the best of our ability. The target of the amplifier system is to achieve a pass-band gain of 25 times the input voltage. This pass-band has cut-off frequencies of 20 Hz and 40 kHz[p. 20, 3]. The input signal is expected to have a maximum amplitude of 0.4 volts. These are just some of the requirements for the amplifier system; a full list of requirements is listed in the chapter 3 (Project Activities).

The power supply must be designed to provide a positive and a negative voltage. The unloaded voltage should be 20-22 V DC, while maintaining an output voltage of 17-20 V DC (with a maximum of 5% ripple) when delivering 1.0 A.[p. 18, 3]

The audio amplifier filters must have a frequency range of 20 Hz to 20 kHz and the different frequency ranges supplement each other well and the transition in frequencies between the different types of filters is as smooth as possible. It is also stated that the amplitude and phase of the different frequency ranges are correct in a way that the total transfer is as flat as possible. [p.19, 3]

If all primary goals are achieved early, an additional challenge is to incorporate a Booming Bass Extension alongside the main amplifier system to further enhance the low-frequency response around 20 Hz. The designing, synthesizing, and implementation of these distinct components will use the knowledge obtained earlier in the Linear Circuits A and B courses.

3

Project Activities

3.1. Tasks Required

To achieve the set goal (See chapter 2), it is necessary to design and build an adequate power supply, several filters for the speaker cones and a power amplifier. This will be done by several sub-groups, each having their respective tasks and requirements, but will help the others when it is required and when they have the opportunity and time to do so.

3.2. Meeting Functions

Table 3.1: Functions during meetings

Week #:	Chair	Secretary
Week 2.2		
19/11/2024	T.J. Bello	K.W. Smid
21/11/2024	S.J.H. ter Huurne	N.T. Herijgers
Week 2.3		
26/11/2024	N.T. Herijgers	K.W. Smid
28/11/2024	S.J.H. ter Huurne	T.J. Bello
Week 2.4		
03/12/2024	N.F. van Vroonhoven	T.V. Wever
05/12/2024	T.V. Wever	N.F. van Vroonhoven
Week 2.6		
17/12/2024	A. Guo	G.D. van Weelden
19/12/2024	G.D. van Weelden	A. Guo
Week 2.7		
07/01/2025	I. Givony	L.D. Melissant
09/01/2025	L.D. Melissant	I. Givony
Week 2.7		
14/01/2025	Repair	Repair
16/01/2025	Repair	Repair

The persons who will fulfill the functions of chair and secretary in the various meetings can be seen in table 3.1.

3.3. Work Breakdown

A small breakdown and overview of the work and tasks each subgroup has:

Table 3.2: Power-Supply Work Breakdown

Power Supply

- **Integrated Project-1 Manual**
Analyzing and dimensioning the power supply, calculating what the value of the capacitor needs to be for a ripple voltage smaller than 5%. The chosen capacitance, between $4700\mu\text{F}$ and $6800\mu\text{F}$, should be such that the ripple voltage does not exceed this threshold.
 - **LTSpice**
Simulating the power supply in LTSpice using the calculated values of the capacitor and building the power supply. The simulations should make it clear whether the calculated value of the capacitor ensures that the ripple voltage remains under 5%.
 - **Power-Supply Circuitry**
Measuring the characteristics of the built power supply and analyzing the data gained. Measurements from the built power supply will be then compared to the simulations to see whether the values and results are expected in reality.
-

Table 3.3: Power-Amplifier Work Breakdown

Power Amplifier

- **Integrated Project-1 Manual**
Analyzing & designing the Power Amplifier circuit and determining values. These values include the resistors: R_a , R_b , and R_c , and the capacitors: C_1^* and C_X . Analyzing the Power Amplifier circuit should make clear what the values of each of these components are and whether they are needed in the first place.
 - **LTSpice**
Simulating the Power Amplifier circuit in LTSpice and building the Power Amplifier. The simulations should make it clear whether the choices made before, which values for the resistors and capacitors or even using them, are desired for the expected results.
 - **Power-Amplifier Circuitry**
Measuring the characteristics of the built Power Amplifier and analyzing the data gained. The measurements should give insight into whether the goal for the Power Amplifier is achieved: non-inverting configuration, voltage gain of 25 and amplification of any DC offset at the input by 1.
-

Table 3.4: Low-Pass Work Breakdown

Bass / Low Pass Filter

- **Integrated Project-1 Manual**
Calculating the values of the Low-Pass components and deciding whether a First-order Low-pass Filter (FO-LPF) or Second-order Low-pass Filter (SO-LPF) and thus a Zobel network should be implemented into the design.
 - **LTSpice**
Designing the said circuit with the insight gained from calculations, simulating the circuit in LTSpice and plotting them in Python. Results should give insight into whether decisions make the filter viable or whether changes are needed.
 - **Low-Pass Circuitry**
Building the circuit and measuring the characteristics of the built Low-Pass Circuit and analyzing the data gained. Seeing if the circuit is working and the measurements should show if the filter is working as intended.
 - **Band-Pass Circuitry**
Adjustments to the Low-Pass circuitry if needed.
-

Table 3.5: Band-Pass Work Breakdown

Mid Toner / Band-Pass

- **Integrated Project-1 Manual**
Doing a speaker analysis and calculate the values of the Band-Pass components and deciding whether a First-, Second-, Third- or Fourth-Order filter is needed to achieve the desired frequency range and the detement transition between the Low- and High-Pass filter and determine whether a Zodal network consisting of a capacitor and an resistor is beneficial for the filter.
 - **LTSpice**
Designing the said circuit with the insight gained from calculations, simulating the circuit in LTSpice and plotting them in Python. Results should give insight into whether decisions make the filter viable or whether changes are needed.
 - **Band-Pass Circuitry**
Building the determent circuit and measuring the characteristics of the built Band-Pass Circuit and analyzing the data gained based on theoretical data we calculated before.
 - **Band-Pass Circuitry**
Adjustments to the Band-Pass circuitry if needed.
-

Table 3.6: High-Pass Work Breakdown

Tweeter / High-Pass
• Integrated Project-1 Manual Calculating the values of the Low-Pass components and deciding whether a First-order High-pass Filter (FO-HPF) or Second-order High-pass Filter (SO-HPF) and thus a Zobel network should be implemented into the design. • LTSpice Designing the said circuit with the insight gained from calculations, simulating the circuit in LTSpice and plotting them in Python. Results should give insight into whether decisions make the filter viable or whether changes are needed. • Band-Pass Circuitry Building the circuit and measuring the characteristics of the built High-Pass Circuit and analyzing the data gained. Seeing if the circuit is working and the measurements should show if the filter is working as intended. • High-Pass Circuitry Adjustments to the High-Pass circuitry if needed.

3.4. Project limits

This Integrated Project aims to build an electrical system to drive a speaker box. This system should have the following characteristics:

- The amplification of a tone should only depend on the input amplitude and not on its frequency.
- The frequency of the audio system range shall be between 20 Hz and 20 kHz (-3 dB bandwidths). It may only send filtered amplified sound signals between 20 Hz and 20 kHz to the speaker.
- The output voltage to the speakers can be 10 V at maximum.
- The signal source produces an electrical signal with an amplitude of a maximum of 0.4 V.
- The system should be put into the IP-1 chassis.

Organization & Schedule

4.1. Organization & Schedule

An overview of how the subgroups are built up can be seen in table 4.1.

Table 4.1: Subgroups

Subgroup:	Students:
Power Supply:	A. Guo & G.D. van Weelden
Power Amplifier:	I. Givony & L.D. Melissant
Low Pass Filter:	N.T. Herijgers & K.W. Smid
Band Pass Filter:	T.J. Bello & S.J.H. ter Huurne
High Pass Filter:	T.V. Wever & N.F. van Vroonhoven

The following table contains the schedule and planning for the upcoming weeks:

Table 4.2: Group Schedule

Week #:	Tuesday	Thursday
Week 2.1	Attending LTSpice CourseLabs	Attending LTSpice CourseLabs
Week 2.2 18 - 22 Nov.	Attending Group Meeting & Course-Labs The first group meeting will emphasize establishing the structure for the Booming-Bass project. This will involve mapping out the key tasks for each subgroup and group assignments to be completed in the upcoming weeks.	Attending Group Meeting & Course-Labs Second group meeting.

Week #:	Tuesday	Thursday
Week 2.3 25 - 29 Nov.	Deadlines - Tuesday 26 Dec. (15:00) PPD - Project Plan Draft TCC - Team Commitment Contract - Wednesday 27 Dec. (22:00) BuddyCheck Buddycheck Review 1 Group Meeting & CourseLabs Mentor Ethics Assignment & discussion.	- Thursday (13:45 - 14:15) Oberservation - Group Observation. Attendance is necessary.
Week 2.4 1 - 5 Dec.	Deadlines - Monday 2 Dec. (15:00) PPD - Project Plan Final Group Meeting & CourseLabs Every subgroup should have started their simulations in LTSpice.	Group Meeting & CourseLabs Every subgroup should have completed their LTSpice simulations, gained measurements and graphs and analyzed them for their intermediate report.
Week 2.6 15 - 20 Dec.	Deadlines - Wednesday 17 Dec. (22:00) BuddyCheck - Buddycheck Review 2 - Friday 20 Dec. (22:00) Report - Intermediate Report Group Meeting & CourseLabs Every subgroup should have started building their circuit and gaining measurements from reality.	Group Meeting & CourseLabs Every subgroup should have completed the Integrated Project-1 Manual and have built their circuitry, since the deadline for the intermediate report is this Friday at 22:00.
Week 2.7 6 - 10 Jan.	Group Meeting & CourseLabs Working on the Final Group Report.	Group Meeting & CourseLabs Working on the Final Group Report.
Week 2.8 13 - 17 Jan.	Deadlines - Friday 17. Jan (22:00) Group Report - Final Group Report Group Meeting & CourseLabs Working on the Final Group Report.	Group Meeting & CourseLabs Finishing the Final Group report.
Week 2.9 20 - 24 Jan.	Deadlines - Monday 20. Jan (22:00) BuddyCheck - Buddycheck Review 3 - Tuesday to Friday Exam - Group & indiv. oral exam No CourseLabs	No CourseLabs

Subgroup Schedule

Presented below are the planning and schedules for each subgroup. In addition to the outlined plans and schedules, are also included a small summary of completed tasks within the subgroup:

Table 4.3: Power-Supply Schedule

Week #:	Tuesday	Thursday
Week 2.2	Integrated Project-1 Manual Analyzing and dimensioning the power supply: deciding which size, $4700\mu\text{F}$ and $6800\mu\text{F}$, of the capacitor is sufficient for the power supply. Doing the simulations of the said power supply with both sizes of capacitors to check whether calculations were approximately correct in LTSpice and gaining measurements for later discussion.	Integrated Project-1 Manual Building and measuring the power supply with the given capacitor that approximates the calculations and comparing the measurements gained from reality to the LTSpice simulations.
Week 2.3	Overleaf & Documentation Appending all LTSpice simulations gained from the previous weeks, graphs, and measurements: therefore all the results gained from building the power supply with the chosen size of the capacitor.	Open to planning & discussion. May provide assistance & support to other subgroups.
Week 2.4	Open to planning & discussion. May provide assistance & support to other subgroups.	Open to planning & discussion. May provide assistance & support to other subgroups.

Table 4.4: Power-Amplifier Schedule

Week #:	Tuesday	Thursday
Week 2.2	Background knowledge Practicing with given exercises of power amplifiers to get an understanding of what we will have to calculate in week 2.3.	Background knowledge Understanding what a power amplifier is, what our end goals for the power amplifier and how to reach them.
Week 2.3	Integrated Project-1 Manual Analyzing & designing the Power Amplifier circuit and determining values. These values include the resistors: R_a , R_b , and R_c , and the capacitors: C_1^* and C_X .	Integrated Project-1 Manual Simulating the Power Amplifier circuit in LTSpice and building the Power Amplifier.
Week 2.4	Integrated Project-1 Manual Measuring the characteristics of the built Power Amplifier and analyzing the data gained.	Overleaf & Documentation 1. Last measurements of the characteristics of the built Power Amplifier and analyzing the data gained. 2. Appending LTSpice simulations, graphs and measurements: therefore all the results gained from building the power supply.

Table 4.5: Low-Pass Schedule

Week #:	Tuesday	Thursday
Week 2.3	Integrated Project-1 Manual Calculating the values of the Low-Pass components and deciding whether a FO-LPF or SO-LPF should be implemented into the design.	Integrated Project-1 Manual Continuing the calculations needed and whether a FO-LPF or SO-LPF should be implemented.
Week 2.4	Integrated Project-1 Manual Designing the said circuit with the insight gained from calculations, simulating the circuit in LTSpice and plotting them in Python.	Integrated Project-1 Manual Analyzing the results gained from LTSpice and Python plotting and deciding what type of filter should be utilized.
Week 2.6	Integrated Project-1 Manual Building the circuit and measuring the characteristics of the built Low-Pass circuit and analyzing the data gained. Seeing if the circuit is working and the measurements should show if the filter is working as intended.	Integrated Project-1 Manual Overleaf & Documentation Adjustments to the Low-Pass circuitry if needed.

Table 4.6: Band-Pass Schedule

Week #:	Tuesday	Thursday
Week 2.3	Integrated Project-1 Manual Calculating Mid-Range speaker values and the Band-Pass component values needed.	Integrated Project-1 Manual Continuing the calculations needed for the circuitry and attempting to finalize the chosen component values that will be used for the simulations.
Week 2.4	Integrated Project-1 Manual Designing said circuit with the gained values of each component, and simulating the circuit in LTSpice. Last minute adjustment to component values will be made if simulation results aren't as expected.	Integrated Project-1 Manual Physically building the circuit and measuring the characteristics of the built Band-Pass circuit and analyzing the data to see if they match the expected results.
Week 2.6	Integrated Project-1 Manual Continuing building the circuit and gaining measurements from it. Analyzing the data and measurements gained to compare them to the LTSpice simulations.	Integrated Project-1 Manual Overleaf & Documentation Adjustments to the Band-Pass circuitry if needed.

Table 4.7: High-Pass Schedule

Week #:	Tuesday	Thursday
Week 2.3	Integrated Project-1 Manual Calculating the values of the Low-Pass components and deciding whether a FO-HPF or SO-HPF should be implemented into the design.	Integrated Project-1 Manual Continuing the calculations needed and whether a FO-LPF or SO-LPF should be implemented.
Week 2.4	Integrated Project-1 Manual Designing the said circuit with the insight gained from calculations, simulating the circuit in LTSpice and plotting them in Python.	Integrated Project-1 Manual Analyzing the results gained from LTSpice and Python plotting and deciding what type of filter should be utilized.
Week 2.6	Integrated Project-1 Manual Building the circuit and measuring the characteristics of the built High-Pass circuit and analyzing the data gained. Seeing if the circuit is working and the measurements should show if the filter is working as intended.	Integrated Project-1 Manual Overleaf & Documentation Adjustments to the High-Pass circuitry if needed.

5

Risks Analysis

5.1. Project Risks and Precautions

During this project, some risks can be expected which could impact the project or the planning of the project. These risks should be addressed beforehand and prevented during the project.

5.1.1. Physical Risks

When limiting physical risks, it all comes down to paying attention and being careful about your work. Obviously, it was agreed that all should be careful and check each other's work to prevent injuries. That being said, we made some guidelines on how to properly prevent the most likely injuries.

Cuts and Bruises Due to Tool Use

During this project, tools will be used and operations will be carried out that carry a certain degree of risk with them. Any type of tool has some risk of serious injury. Therefore, certain guidelines were made on the use of hand and power tools to try to minimize the risk of injuries.

1. **Check.** Check the state of the tool and see if it is safe for use or if it requires maintenance and cannot be used at all.
2. **Plan.** Plan what actions you will be performing with the tool and plan how it will work. See if you can reach all the areas you need to reach and if there are no obstructions to your work.
3. **Execute.** Pay attention to your work while performing the tasks. Tools are nothing to joke with, and their use should always be taken seriously.

Burns

During the project, certain things have to be soldered. Since soldering irons get very hot, this poses a serious risk of burning. The rules for tool use still apply for soldering irons, but because of the added risk, there are some extra precautions to be taken. Mainly:

- Clean your work environment and make sure you stay aware of where the soldering iron is, as well as making sure everyone in the work area is aware of the fact that the soldering iron is hot.
- After and in between usage, make sure to put the iron in the appropriate holder, as well as making sure the iron is turned off when not in active use.

Electrocution

During the project, the speaker will be powered by electricity. Since the power is supplied by very strong power supplies, electrocution is a risk. Preventing this is once again a case of checking our work and staying aware of live sources. We also agreed to keep conductive materials and liquids, such as water or laptops, away from live wires.

Injuries Due to Fights

Preventing fights is a prime example of keeping relations in check. There are many ways relations can suffer and therefore cause fights or other issues. Some known reasons for frustration are discussed in the TCC. Therefore, by agreeing to and following the TCC, fights are being prevented.

Limited components

There is a risk involved in only a limited amount of components being available, and the group only getting a limited amount of them. This poses a restriction on the filters. For example, the filters are only allowed to contain two capacitors and two inductors. This means that these groups must be careful with their decisions, as they may waste components otherwise. This means that there is a risk involved with picking the right components.

Communication

All groups work together closely, ensuring that all the components work as they should do. For example having the right phases, voltages and currents. If there is bad communication between subgroups, it could put the end result of the project at risk. Therefore it is vital that all subgroups work together and communicate effectively.

6

Conclusion

In conclusion, this project plan will be followed to help overcome challenges that may arise. As outlined in Chapter 2 (Project Results), the specifications and requirements for the loudspeaker are clearly defined, while the project limits and constraints are detailed in Chapter 3 (Project Activities). The schedules of each subgroup will serve as the foundation for this approach, ensuring effective coordination and minimizing the risk of miscommunication between subgroups. Additionally, the work breakdown provides a clear delineation of responsibilities and tasks for each subgroup.

Power Supply

The power supply group successfully designed and built their power supply. The specifications were that the maximum voltage output was 22 V, that the load current was 1 A, with an average output of 17 - 20 V, and lastly that the ripple voltage did not exceed 5%. The power supply was designed in a way that this was achieved successfully, with validation provided through simulations conducted using LTSpice, and measurements.

Power Amplifier

The power amplifier group was able to design and build an amplifier that met the required specifications. Ideally, this power amplifier can suppress all frequencies below 20 Hz and above 40 kHz. That is done with the low pass filter and high pass filter before the input of the opamp in the circuit. The pass-band gain of this circuit is 25. This means that the voltage at the input is amplified 25 times, which is equal to an added gain of about 28 dB to the input 'volume' of 0 dB. Furthermore, any DC offset at the input of the amplifier is amplified by a maximum of 1, to suppress any clipping of the speaker. The eventual range of frequencies is about 20 Hz - 41 kHz with a gain of about 28.2 dB (according to simulations). This means that at least the design of the power amplifier was successful.

Filters

The three filter groups all worked together to get to an end product. First, the speaker data was analyzed with the help of Python. A model of the loudspeaker was created. Using the data, along with the model, the value of the impedance in the loudspeaker model was created. This created a base for the work on the filters. After that, the acoustic data of the speaker was analyzed cooperatively. This acoustic data indicates how the speaker will react at each frequency and what the power of the sound that it generates, is. This led to a decision about the cut-off frequencies, so what frequencies each speaker will produce.

Low-Pass

It was found that the **low-pass** filter had to take the cut-off frequency at 150 Hz. At this point, this filter will reduce its power output to less than 50% of its peak value. Python was used to simulate the circuit and to find the optimal match for the components of the filter attached to the loudspeaker. This led to a match with the available components and a cut-off frequency of 150 Hertz.

Band-Pass

After that data analysis of the loudspeaker, it was decided that a second-order filter should be used for the **band-pass** filter. It was inferred from the acoustic data analysis that the cut-off frequencies should be 150 Hz and 1250 Hz. This means that the band-pass filter will start accepting frequencies between 150 and 1250 Hz, and reject anything outside of this. Python was used again to find the best match for the combination of components.

High-Pass

For the **high-pass** filter, similar measures were taken. After collecting data on their loudspeaker impedance, it was discussed with the other groups how to find the cut-off frequencies for each filter. The conclusion was that it may be appropriate for the high-pass filter to have an earlier cut-off frequency than the band-pass filter. This will allow for a smoother transition from band-pass to high-pass, as the mid-toner is sensitive around the cut-off frequencies mentioned before. This raised questions about the phase, as the band-pass filter must be in phase with the high-pass filter at the cut-off frequencies. When simulations were performed, the high-pass filter was successfully designed and synthesized, with the appropriate cut-off frequency.

Team Commitment Contract

Team name: B2-1

Shared team values:

1. **Be present (Physically & mentally).**
- We discussed the importance of being physically present during our team meeting. We all agreed that being physically there will increase our performance, as well as allow us to communicate better. Therefore we put it in the contract, to increase our likelihood of success.
2. **Keep your promises.**
- During the team meeting we agreed that by making promises to each other regarding the work, deadlines or anything regarding the project, we will make sure to keep them. This is important to us as it will show that each of us is committed to the project, and it will keep any false promises from happening.
3. **Be direct but respectful.**
- We think that the ability to be direct with one another is an important thing, as it will create an environment where open communication can happen. This will be important during the project, as we can then give one another useful tips, or feedback. Additionally, we agreed that being respectful is another key factor, as it will keep open communication from being hurtful towards others.
4. **Integrity.**
- We all agreed that having integrity is a very important thing in this project. Because we are continuously working together, we agreed that we should all keep the same moral standard in our work, so no copying or using artificial intelligence as this will harm the academic integrity of our work.

Target or ambition level:

What grade are you working for?

- We are all aiming for a 10.0. This is a high grade, but we agreed that setting our goals and expectations high will motivate us to work harder and reach this goal. Even if we do not all reach this grade, striving for a 10.0 will allow us to work to our maximum potential.

Quality:

How will you prevent fraud and plagiarism?

- By regularly asking each other to check our work. By doing this we will not only make sure that the quality of the work is of a high standard, it also ensures that the group mates can spot and point out any potential plagiarism. Moreover, we instilled a policy where we will not copy anything from others or the internet, such as AI. This can therefore help us to prevent plagiarism

How you will organize feedback (relation, process, content)?

- We have two team meetings a week, one on Tuesday and one on Thursday. We will dedicate some time during every team meeting on Thursday for feedback. We will allow people to give their feedback on their sub-group's work and the whole group's work.

How do you determine the quality of the work your group and each team member does?

- During the project, other group members will read over everyone's work, allowing them to spot and point out any mistakes. Moreover, we will check sources as there are a lot of unreliable sources that can harm the quality of our work. Therefore having quality sources can help to raise the quality of our work. Lastly, we will run simulations when possible, we have tools such as 'LT Spice' or 'Python' available to us, which can help us determine whether the work is up to the standards we have set, using the knowledge we have about the subject.

Workload:

How do you make sure that the workload is fairly and evenly distributed?

- We decided in a team meeting that this is up to the sub-groups to decide, as the sub-groups are most aware of their workload. This therefore puts the responsibility of an equitable workload on the people in their respective sub-groups. Additionally, we decided that if other group members are aware of the workload not being distributed fairly in a specific sub-group, then this can be called out and action can be taken. This may include a discussion in the next team meeting, and distributing the workload more fairly.

Planning:

How do you ensure that each team member finishes everything on time? Did you clarify who will have a final say in the final deliverable and submit it to Brightspace on behalf of the project group?

- We also decided that the responsibility of finishing things on time first lies within the sub-groups, however, if deadlines are not being met then the other group members will start to get involved, as they are directly affected by the slacking behavior of other group members. We have decided on a group member who will deliver the final project on Brightspace.

Behavior:

How do you treat each other in the group?

- The most important thing is to treat each other as equals and with respect. By treating each other as equals, we can introduce new ways of thinking into the group and stimulate a safe working environment. By always being respectful we can ensure that we avoid conflicts and maintain a good place for working.

How do you handle disagreements/conflicts within your group?

- Disagreements are bound to happen, as we are always working on the same project together, and therefore it is very likely that there will be differing opinions. We want to handle these conflicts in an orderly fashion, first allowing the disagreeing people to talk about why and find some sort of middle ground. If this is not possible there will be a third member involved, who can then help find a middle ground.

Could your guide or TA be involved in reaching consent?

- If there is a conflict, and there has already been a third member involved without being able to find a middle ground, we could ask a TA or guide to help find a middle ground. For inspiration we may also be able to ask TA's for help, however, we do not want the TA's to introduce ideas that we would not

have been able to come up with ourselves, as that might be considered inequitable.

What do you do if someone is late during a group meeting?

- We will first encourage them to explain why they are late, as they might have a valid reason that caused them to be late. If this reason is not valid we will give them a warning. If they continue to be late we will sit down as a group and discuss it in detail. We will explain the harm it does to the team to encourage the person not to be late again. If this does keep happening, the late comer could be assigned any task this person wasn't present for when the group was discussing who is responsible.

Communication:

In what ways do you communicate with each other as a group and among yourselves? Zoom, MS Teams?

- We will communicate through WhatsApp as this is where we are all most active, which will allow us to answer questions that others have as quickly and as often as possible. Moreover, we will be using Notion for the planning of the project, it has a built-in calendar and a task divider, which will make it clear what each person has to do both daily and weekly.

What information do you share via WhatsApp, e-mail, telephone?

- We will share most of our information through WhatsApp, as this is the quickest form of communication for all of us. This might include calculations, text for in any group document or Python code. Otherwise, we might use E-Mail for files and Notion for task division and planning.

What are your office hours?

- We decided that we want the weekdays, Monday to Friday, from 08:30 to 18:00 to be the 'Office Hours'. This is when we should expect questions to be answered quickly, and answer questions ourselves quickly too. During the 'Office Hours' we should all be as active as possible. After 'Office Hours' we will not be expected to answer questions, however, we still can, as we will likely work on documents for the project at the weekends. Team members should understand and respect that someone may not respond after the 'Office Hours' as they have other things to do as well.

How do you become an expert on the topic you are not directly involved in?

- During the team meetings on Thursday, when we do an update on each of the sub-groups, we expect them to give a detailed description of anything they are working on and explain how it works. This allows the other group members to understand how other components work and also become experts.

Decision-making:

How do you make decisions? By majority vote or by consensus?

- We will make our decisions by consensus most of the time, as this will please most of the group. It will not be likely that there are too many differing opinions as we all agree on most things. Therefore consensus can give us the best results.

Consequences:

What are the consequences if a participant in the group does not keep the agreements?

- We will create a strike agreement, if someone gets three strikes they will be expected to bring cake the next day and share it with our group. This is a fun way of making sure that people will be keeping their agreements. Because this approach may not seem too serious we will also make sure that at three strikes, the person who has the strikes will get a serious talking to from the other group members. They will explain the consequences of not keeping the agreements.

Success factors:

What makes your team a dream team?

- What makes us a dream team is that everyone in the group is motivated to work hard and is eager to learn. This creates an optimal environment for group work and allows us to become a good team and excel in the project.

Norms or evaluation criteria

You will evaluate your own and each other's work in this project.
Discuss and write down your team criteria. You need at least five different criteria.

1. Keeps deadlines
2. Writes everything themselves
3. Being able to explain what they worked on in detail
4. How well they participated throughout the project
5. The quality of the work

Signature of involved team members and date:

Niek F. van Vroonhoven | 24 - 11 - 2024

Alex Guo | 24 - 11 - 2024

Stijn J.H. ter Huurne | 25 - 11 - 2024

Karsten W. Smid | 25 - 11 - 2024

Noah T. Herijgers | 25 - 11 - 2024

G.D. van Weelden | 25 - 11 - 2024

Itai Givony | 27 - 11 - 2024

Teun J. Bello | 28 - 11 - 2024

Tycho V. Wever | 28 - 11 - 2024

Luuk D. Melissant | 01 - 12 - 2024

Bibliography

- [1] Charles Alexander and Matthew Sadiku. *Fundamentals of Electric Circuits*. 7th ed. McGraw-Hill Education, 2021.
- [2] Roel Grit. *Project Management: A Practical Approach*. 4th ed. Groningen, The Netherlands: Noordhoff Uitgevers, 2015.
- [3] G.J.M. Janssen et al. *Student Manual: Integrated Project 1 (IP-1) EE1L1 'Booming Bass (Sound) System'*. Tech. rep. 2024.